

Recursive Formulas: Sequences and Terms

Directions: Show all work. No calculators.

1. Write a recursive formula to represent the sequence 6, 11, 16, 21, ...
2. Write a recursive formula to represent the sequence 32, 25, 18, 11, ...
3. Write a recursive formula to represent the sequence $-12, -4, 4, 12, \dots$
4. Write a recursive formula to represent the sequence 17, 12, 7, 2, ...

5. Write a recursive formula to represent the sequence $-5, -11, -17, -23, \dots$

6. Write a recursive formula to represent the sequence $3, 15, 27, 39, \dots$

7. Write a recursive formula to represent the sequence $41, 31, 21, 11, \dots$

8. Write the first 6 terms of a sequence if $a_1 = 18$ and $a_n = a_{n-1} + 7$.

9. Write the first 6 terms of a sequence if $a_1 = -4$ and $a_n = a_{n-1} + 9$.

10. Write the first 6 terms of a sequence if $a_1 = 35$ and $a_n = a_{n-1} - 6$.

11. Write the first 6 terms of a sequence if $a_1 = -20$ and $a_n = a_{n-1} + 5$.

12. Write the first 6 terms of a sequence if $a_1 = 7$ and $a_n = a_{n-1} - 8$.

13. Write the first 6 terms of a sequence if $a_1 = 12$ and $a_n = a_{n-1} + 11$.

14. Write the first 6 terms of a sequence if $a_1 = 50$ and $a_n = a_{n-1} - 13$.

15. A sequence has recursive formula $a_n = a_{n-1} + 4$ with $a_1 = -7$. What is the common difference for the sequence? Write the first 5 terms of the sequence.

16. A sequence has recursive formula $a_n = a_{n-1} - 3$ with $a_1 = 26$. What is the common difference for the sequence? Write the first 5 terms of the sequence.

17. A sequence has recursive formula $a_n = a_{n-1} + 12$ with $a_1 = -15$. What is the common difference for the sequence? Write the first 5 terms of the sequence.
18. A sequence has recursive formula $a_n = a_{n-1} - 7$ with $a_1 = 9$. What is the common difference for the sequence? Write the first 5 terms of the sequence.
19. A sequence has recursive formula $a_n = a_{n-1} + 6$ with $a_1 = -22$. What is the common difference for the sequence? Write the first 5 terms of the sequence.
20. A sequence has recursive formula $a_n = a_{n-1} - 9$ with $a_1 = 40$. What is the common difference for the sequence? Write the first 5 terms of the sequence.
21. A sequence has recursive formula $a_n = a_{n-1} + 14$ with $a_1 = 5$. What is the common difference for the sequence? Write the first 5 terms of the sequence.